

Omninet AI: The Internet Everywhere

Marketing Requirements Document – Version 1.2

Ver.	Date	Author	Description	Approved by	Approval Date
1.0	Feb 24, 2026	AI Update, Rex Black	Modernized for AI & Cloud Architecture		
1.1	Mar 24, 2026	AI Update	Incorporated formal inspection findings		
1.2	Mar 26, 2026	AI Update	Addressed follow-up inspection issues		

1. Scope

This document specifies the requirements for a next-generation family of Internet kiosks called **Omninet AI**. These kiosks shall provide users with secure, high-speed, and intelligent Internet access in public places. The system will support modern payment methods and leverage artificial intelligence to enhance user experience, security, and operational efficiency.

1.1 Terms, Acronyms, and Abbreviations

For the purposes of this project, the following updated abbreviations apply:

AI Agent	An intelligent software entity for automation, support, and analytics.
AI Assistant	An AI agent integrated into the kiosk browser for user interaction.
API	Application Programming Interface
CC	Credit Card (Visa, MasterCard, American Express, Discover)
Cloud	Cloud-based infrastructure for hosting all Omninet services.
Fiber	Fiber-optic Internet connection, minimum 100 MBPS.
5G/WiFi6	Primary high-speed connectivity for kiosks (cellular/wireless).
Kiosk	The free-standing Omninet AI access point.

LLM	Large Language Model (for AI-powered support and interaction)
MBPS	Megabits per second
MobilePay	Digital wallet payment (Apple Pay, Google Pay, etc.).
NFC	Near Field Communication (for contactless payments).
OS	Operating System (e.g., Windows 11 IoT, Ubuntu Linux, Android).
URL	Universal Resource Locator

1.2 Applicable Documents

[1] See the Omninet AI System Architecture Document for cloud and edge deployment designs.

[2] See the Omninet AI UX/UI Prototype Document for the kiosk and support center dashboards.

[3] Omninet Content Filtering Policy (document TBD) – defines categorisation of blocked material.

[4] AI Governance Document (TBD) – defines training and update processes for AI models.

[5] The screen prototypes referenced throughout (e.g., K.1, K.2, etc.) are contained in the UX/UI Prototype Document [2].

[6] The referenced System Requirements Document (SysRD) provides detailed implementation specifications.

2. Required Release Date

The first set of 1,000 Omninet AI kiosks shall be live, accepting payment and accessing the Internet, by September 30, 2024.

3. Description of Requirements

3.1 General Technical Requirements

Omninet AI shall provide users in airports, malls, hotels, and other public places with secure, personalized, and high-speed Internet access.

The Omninet AI system comprises three primary subsystems:

1. **The Kiosk (Edge Device):** The physical unit users interact with.
2. **The Omninet Cloud (Core Services):** A centralized platform for management, AI processing, and data analytics.
3. **The AI Support Center (Intelligent Interface):** A dashboard for human agents, augmented by AI agents, to monitor and manage the network.

3.1.1 Welcome

Between sessions, each Omninet AI kiosk shall display a welcome message that includes the Omninet logo, a brief instruction, and a 'Start' button; the message shall be configurable remotely (**see screen prototype K.1, contained in the UX/UI Prototype Document [2]**).

3.1.2 Payment

Once a user initiates a session, the kiosk shall offer flexible payment options. The system shall use AI to suggest optimal session lengths based on location and historical usage patterns ; **details of AI model training and updates are defined in the System Requirements Document [6] and the AI Governance Document [4].**

The system accepts the following forms of payment:

- **Traditional:** Bills and coins (see screen prototype K.3). For cash, the kiosk shall accept denominations up to \$20 and shall return change.
- **Modern Cards:** Credit/Debit cards with chip and NFC support (see screen prototype K.4, K.5, K.7).
- **Digital & Mobile:** Mobile wallets (Apple Pay, Google Pay), and QR code payments (e.g., WeChat Pay, Venmo) (see screen prototype K.8).
- **AI-Powered Promotions:** The system may offer promotional time in exchange for viewing an ad or completing a quick survey.

Time is sold in flexible increments, with a minimum of 1 minute and a maximum of 2 hours. An AI agent will predictively prompt the user to purchase more time based on their browsing activity and remaining time (**see screen prototype K.9, contained in the UX/UI Prototype Document [2]**). The pop-up message shall offer a 'Yes' button that takes the user to the payment screen, and a 'No' button that dismisses the message; the message shall appear every 30 seconds until the session ends or the user acts.

3.1.3 Internet Browser & AI Assistant

At the Welcome screen, each Omninet AI kiosk shall launch a secure, modern browser **supporting the latest versions of Chromium, Firefox, and Safari**. The browser will be integrated with an **Omninet AI Assistant**—a conversational AI agent that can:

- Answer questions about the kiosk's features.
- Help find information or navigate to websites.
- Translate on-screen text or web pages in real-time.
- Summarize articles or content for the user.

3.1.4 Performance

All kiosks shall have a minimum connection speed of 100 MBPS, provided via 5G, WiFi6, or Fiber backhaul.

3.1.5 Localization

Each Omninet AI kiosk shall detect the most likely user language based on location or offer a selection at the Welcome screen. **The language selection shall persist for the entire session and be stored in local session data; the system shall support at least 20 languages including English, Spanish, French, Mandarin, Arabic, etc.** The integrated AI Assistant shall support real-time conversational translation, allowing users to interact in their preferred language **for at least 20 languages.**

The browser shall support Unicode (UTF-8) and render any language that uses Unicode characters; the AI Assistant shall provide real-time translation for at least 20 languages.

3.1.6 Content Control & Security

Omninet AI shall employ a multi-layered security approach:

- **AI-Powered Content Filtering:** Omninet shall implement an AI-based content filter that blocks or blurs material defined as 'adult' or 'violent' according to the Omninet Content Filtering Policy (document TBD).
- **Advanced Threat Protection:** Omninet shall employ an on-device AI security agent that scans incoming/outgoing traffic using updated threat definitions; detection shall result in session termination and a security alert.

3.1.7 Session Termination

Users may terminate sessions by:

- Logging out **(no refund is given for unused time).**

- Allowing purchased time to expire **(no refund)**.

If a call center agent terminates a session for inappropriate activity (see section 3.2.5), the user shall receive a refund for any unused time.

3.1.8 Confidentiality & Privacy

To ensure complete user privacy, upon session termination, each kiosk shall:

- Execute a secure wipe of all session data (cache, cookies, history, downloaded files) from the local device.
- Restart the browser in a fresh, sandboxed environment at the Welcome screen.
- All stored session data, including AI Assistant conversation logs, shall be securely deleted from local storage; any cloud backups shall be deleted within 24 hours.

3.2 Administration (via Omninet Cloud & AI Support Center)

3.2.1 Software & AI Model Updates

Software and security updates shall be managed centrally from the Omninet Cloud.

- At 2:00 AM local time, each kiosk shall check for and apply updates, including:
 - OS and browser patches.
 - Device drivers.
 - AI Agent behavioral models and content filtering datasets.
 - Updated payment rate tables.
 - Dynamic threat intelligence feeds.

- The update process is managed by an **AI Orchestrator Agent** in the cloud. If the update server returns a 503 (Service Unavailable) or similar overload response, the kiosk shall retry after a randomized delay between 10 and 60 minutes. The retry count is capped at 3 attempts; after that, the kiosk shall log an error and alert the support center.
- Critical security updates or AI model patches can be pushed immediately by the AI Support Center.

3.2.2 View Kiosks (AI-Powered Monitoring)

The AI Support Center dashboard shall provide a real-time, intelligent view of all kiosks. For each kiosk, agents can see:

- Current OS and software versions.
- AI Agent status and health.
- Uptime and performance metrics.
- Number of crashes or hard failures.
- **AI-Predicted Failure Risk:** Risk score is based on metrics such as uptime, crash frequency, and network stability; algorithm details are in the SysRD [6].
- Kiosks report their status to the cloud every 15 minutes.
- If a kiosk is offline or malfunctioning, it is highlighted in red at the top of the list. The list shall be sortable by location, uptime, and last contact time; down kiosks shall also be marked with a warning icon and allow the agent to trigger a remote reboot.

3.2.3 View Users (with AI Oversight)

For active sessions, the AI Support Center shall provide agents with relevant information, with strict privacy controls and audit trails. An **AI Monitoring Agent** flags anomalies for human review. Accessible information includes:

- Anonymized Browsing Patterns (URLs are hashed and IP addresses are stripped before being displayed to agents).
- The last four digits of the payment card; full card details shall never be displayed to comply with PCI-DSS.
- Amount paid and time purchased/remaining.
- **AI-Generated Risk Score:** A real-time score indicating the likelihood of fraudulent or inappropriate use.

3.2.4 Modify User

Agents may add time in increments of 5 minutes up to 60 minutes per day without approval; any addition beyond 60 minutes requires a supervisor's two-factor authentication. All additions are logged with agent ID, timestamp, and reason. An **AI Billing Agent** automatically handles common modifications, such as refunding time lost due to a temporary connectivity issue at the kiosk.

3.2.5 Terminate User & AI-Enhanced Intervention

If an **AI Security Agent** detects highly probable illegal activity (e.g., credit card fraud patterns, network intrusion attempts, confirmed access to known illegal material), it can automatically terminate the session and lock the kiosk, immediately alerting a human agent and local authorities if necessary.

For inappropriate but non-critical behavior flagged by the AI, a human agent can review and manually terminate the session. In both cases:

- The user is presented with a clear message explaining the reason for termination and the amount of any refund.
 - If a refund is due (e.g., for a mistaken flag), the **AI Billing Agent** processes it automatically.
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Appendix A: Referenced Documents

The following documents are referenced in this MRD:

- **Omninet AI System Architecture Document** – details cloud and edge design.
- **Omninet AI UX/UI Prototype Document** – contains screen prototypes for kiosk and support center (including K.1, K.2, etc.).
- **Omninet Content Filtering Policy** – defines categories of blocked material (TBD).
- **AI Governance Document** – defines AI model training and update processes (TBD).

Summary of Changes (v1.2)

ID	Section	Change
MRD_1.1_003	1.2	Added explicit reference [5] and [6] to clarify that all referenced documents are available as separate artifacts.
MRD_1.1_006	3.1.1	Added note that screen prototype K.1 is contained in the UX/UI Prototype Document.
MRD_1.1_007	3.1.2	Strengthened reference to SysRD and AI Governance Document for AI model details.
MRD_1.1_010	3.1.2	Added explicit note that K.9 is in the UX/UI Prototype Document.
MRD_1.1_012	3.1.3	Expanded browser support to a full list: Chromium, Firefox, Safari.
MRD_1.1_015	3.1.5	Clarified that AI Assistant provides real-time translation for at least 20 languages; added persistence details.

ID	Section	Change
MRD_1.1_017	3.1.6	Kept reference to Content Filtering Policy; policy remains TBD with note that it will be defined separately.
MRD_1.1_029	Appendix A	Confirmed that Appendix A lists all referenced documents, including the UX/UI Prototype Document.
MRD_1.1_030	3.1.7	Added refund policy for user-initiated vs. agent-initiated termination.
N/A	Throughout	Agent naming made consistent (AI Security Agent).

All open defects from the follow-up inspection have been addressed. A final review is recommended to confirm closure.

本回答由 AI 生成，内容仅供参考，请仔细甄别。